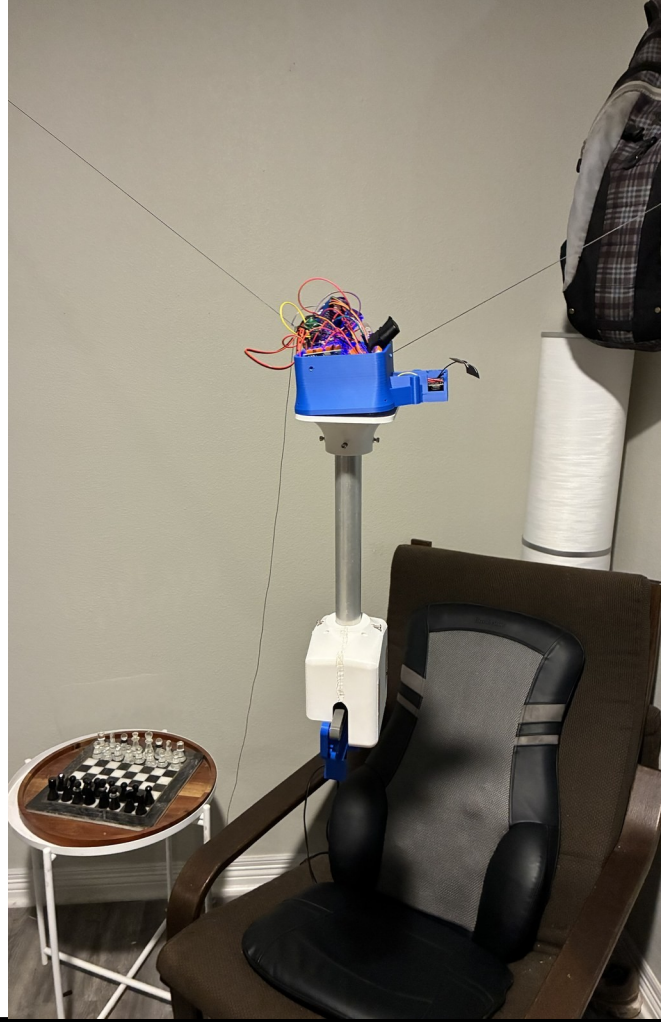
RoomCleaner

A four-cable ceiling robot that picks laundry off the floor

Jasper Buntinx | Mechanical Engineering | Texas A&M

October 27, 2026



About me

Education: Texas A&M

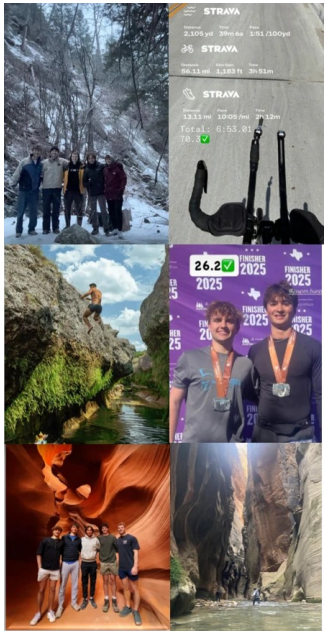
- Bachelor's, Mechanical Engineering, August 2027

Experience

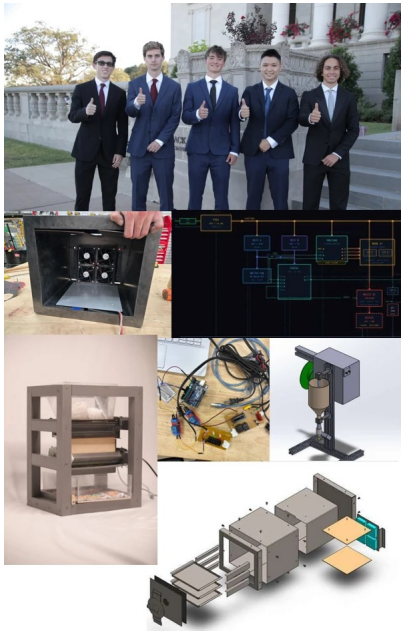
- ITP Lab, undergraduate research
- Rochester Sensors, mechanical intern
- Petrobras, mechanical intern
- Aggies Create, EcoFil electrical lead

Hobbies

- Personal projects
- Soccer
- Running
- Bouldering
- And man, do I love hiking.



Outdoors and endurance



Engineering



Community

Project scope

The problem

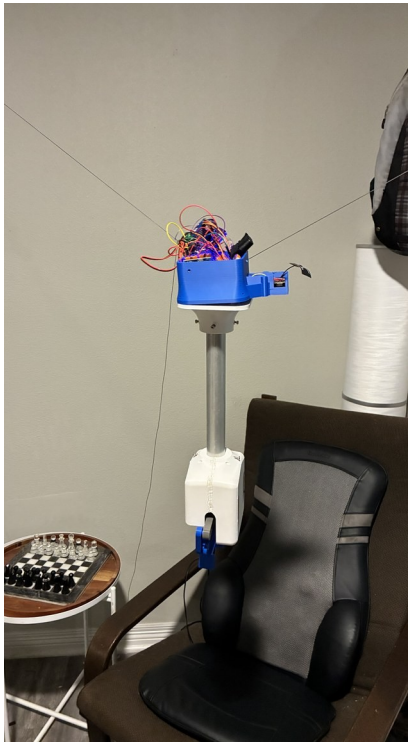
- Laundry on the floor of a 3.35 x 2.74 m room (as reported)
- Ceiling 2.74 m; four studs near the ceiling corners
- Design load: a pair of jeans, 0.9 kg (screening value)
- Plus 2 m/s² for stops and snags (assumed)

The solution

- Four ceiling winches, one claw on four cables
- Nothing on the floor; the room is the frame
- A camera on the claw finds the laundry
- Floor to hamper with no human touch

How we do it

- I approve each move on the phone
- PAUSE holds position with torque on
- The switched power strip is the emergency stop; it drops the claw
- First gate: a 1 lb test weight, then jeans
- Parts in the robot: about $211 for three stations, plus $38 filament (partly estimated)



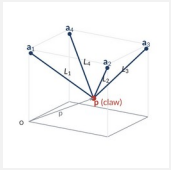
The claw hanging in the room corner (Oct 27)

How cable length becomes position, and torque becomes pull

Cable length → position

$$L_i = \|\mathbf{a}_i - \mathbf{p}\|$$

$\mathbf{a}_i$ anchor, $\mathbf{p}$ claw



- Each spool turn pays out line: 47.1 → 62.8 mm per rev on the Ø15 core as layers build (calculated)
- Four cable lengths, three position unknowns: overdetermined, solved by least squares
- The camera on the floor board fixes the anchors and cable lengths at startup (calibration)
- Validated in the simulator and by held-out poses during calibration (simulated); no physical position measurement yet

Torque → pull on the line

$$I = \frac{V_{\text{ref}}}{8\,R_s} \qquad \tau_{\text{hold}} = \tau_{\text{rated}}\,\frac{I}{I_{\text{rated}}}$$

$$\tau_{\text{run}} \approx 0.7\,\tau_{\text{hold}}\ \text{(assumed)} \qquad F = \frac{\tau}{r}$$

- At 1.5 A: τ_hold 0.36 N·m → running pull 34 N on the empty spool (r 7.5 mm), 25 N on a full spool (r 10 mm) (calculated)
- Validated: Vref measured on the board; the 0.7 running fraction is assumed; the pull test is planned

Pull → what four cables hold

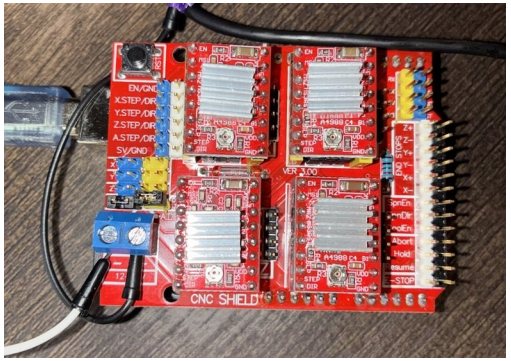
$$T = \frac{W}{4\sin\varphi} \qquad \sin\varphi = \frac{d}{\sqrt{d^2 + D_h^2}}$$

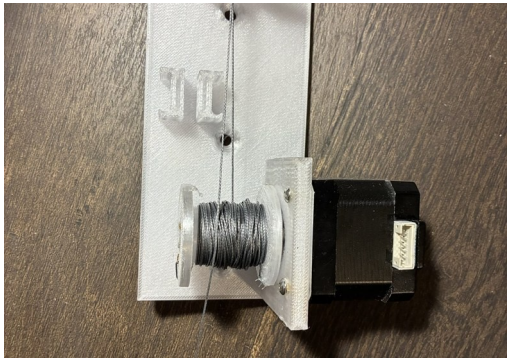
- Tension doubles each time the depth below the anchors halves: 32 / 16 / 8 / 5 N at 0.15 / 0.3 / 0.6 / 1.2 m for 1 lb + claw (calculated)
- Why the planner keeps the claw at least 0.4 m below the anchors
- Validated: calculated only; no cable tension has been measured on the rig

Generation 1: one central unit

The first design

- One Uno and one CNC shield drove four corner motors
- Wires from that one box to every corner
- An overhead ceiling camera scanned the room
- A printed corner plate screwed into a joist: a hook is a pivot, a plate is not
- Each cable found zero against a bead at the outlet
- Bench: a station lifted a test weight; the cable math held (measured)







The central board; a gen-1 corner plate; a gen-1 station lifting a test weight

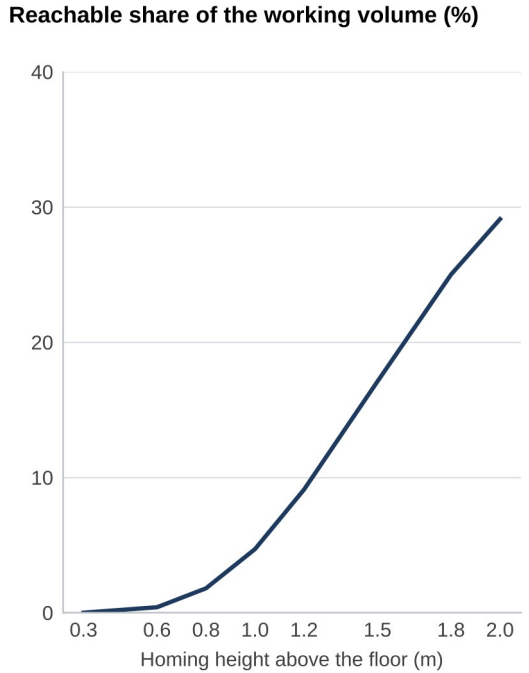
Problems with Generation 1

Wiring and the bench

- Every motor and every sensor wired back to one place
- 18 V on the meter from a supply labelled 12 V (measured)
- A homing switch that would have zeroed without moving: logic inverted
- The switch had to move below the pulley

The architecture

- One board stepping four motors skews them: 2.5 mm per 50 ms (calculated)
- Bead homing: a cable can never be shorter than at home
- So it capped the room at 5-29 % of the working volume (calculated)
- The ceiling camera: one more device per room; the claw and cables hide the grasp spot



Calculated (bead reach model)

What a winch on a wall needs to work

The switch (button)

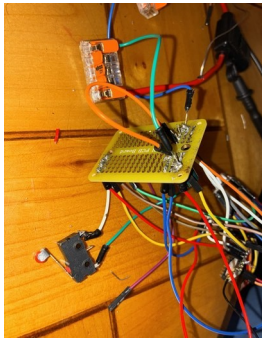
- Things go wrong: a camera fails, a spool overwinds, tension rips a mount out, a motor stalls
- A switch the cable's bead can hit: that station stops
- Fail-safe wiring: LOW at rest, HIGH pressed or with a broken wire

The encoder

- On the spool: measures the discrepancies while testing; data first
- Confirms every commanded step
- Catches slip: the tension sensor
- If the camera proves good enough, it can go later

And now all this wiring, from one box

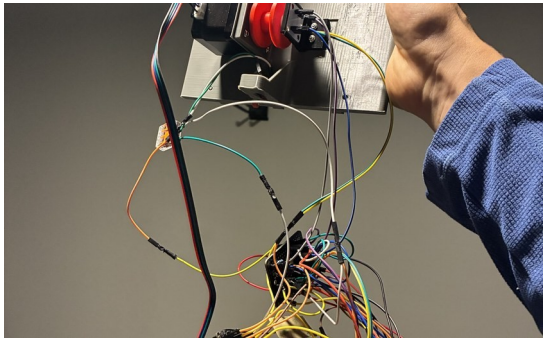
- Motor, switch and encoder in every corner: 10 conductors per corner
- 250 m (820 ft) of wire back to one place (calculated)



Homing switch



Switch pressed into the slide-in



One slide-in, its harness hanging

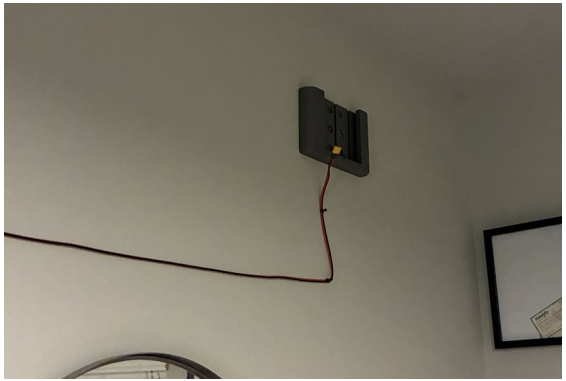
The solution: a modular box that only needs power

A box with its own brain

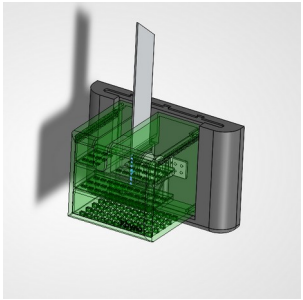
- An ESP32 in each box, with its driver, buck, switch and encoder
- The only thing going in is power: one 12 V pair
- 50 m of wire instead of 250 m (calculated)
- Power can come from anywhere, so a box sits in any position and orientation
- The room's geometry no longer limits where the winches go

Why 16 AWG

- Drop = current x loop resistance
- Longest branch 32 ft one way: 64 ft of loop
- At 2 A: 0.82 V lost on 18 AWG, 0.51 V on 16 AWG (calculated)
- That is 7 % against 4 % of 12 V; less heat in the wall
- Cost: a 100 ft roll about $30; 65 ft used for three stations, about $19.50 (estimated)



The wall mount: one power plug



The station: board, driver, motor, fin (CAD)

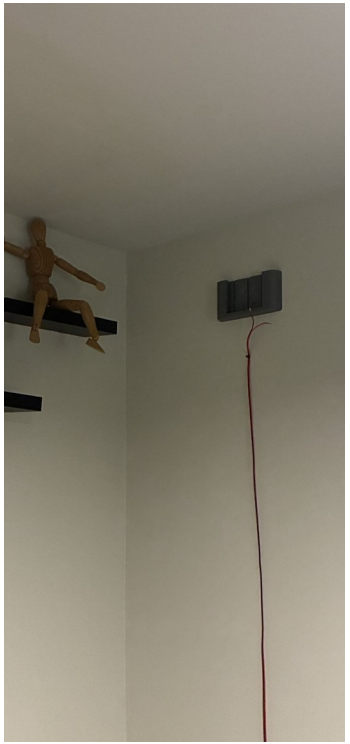
Modular, for the everyday user

Unmount, remount, plug in

- A box slides off its wall mount and unplugs
- Guests coming: each box goes in a drawer
- Back on: one plug, it joins the Wi-Fi by itself

No screwdriver on the wall

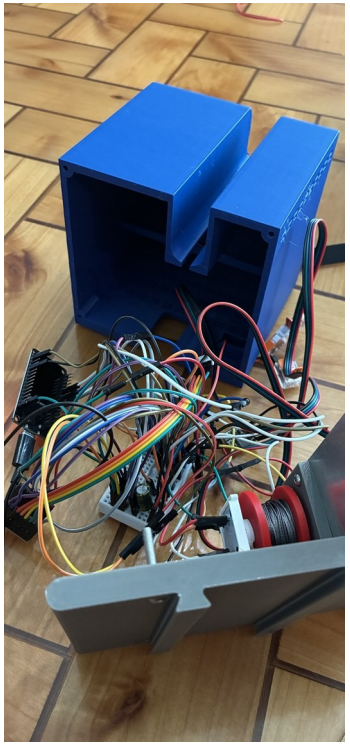
- Slip the station out of the mount, open the electronics
- Close it, slide it back: the mount and its screws stay in the stud
- The mount is separate: a stud today, a pole clamp later



The wall mount alone, lead hanging



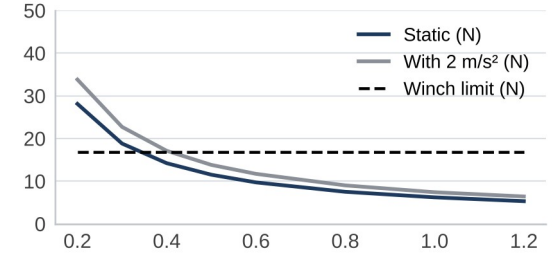
The station slid onto it



Shell open on the bench

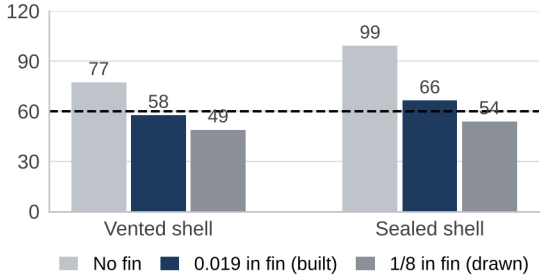
Trade-offs: what each component needs

Tension per cable (N) vs depth (m)



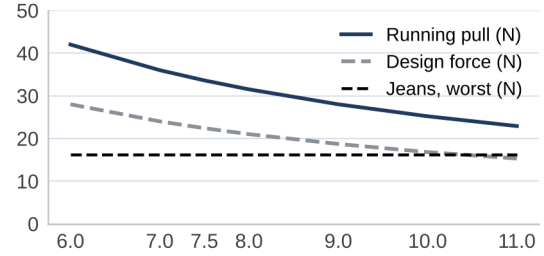
Need: stay 0.4 m below the anchors with 1 lb, 0.6 m with jeans

Motor case (°C) at 1.5 A; PETG limit 60 °C



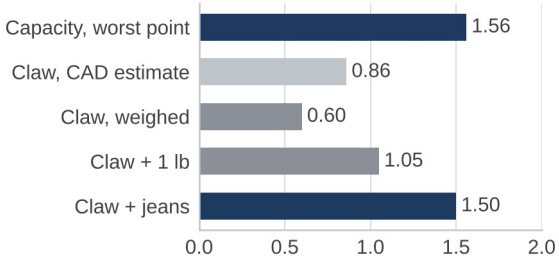
Need: below 60 °C; vented shell + built fin 56-59 °C (calculated)

Pull per winch (N) vs spool radius (mm), 1.5 A



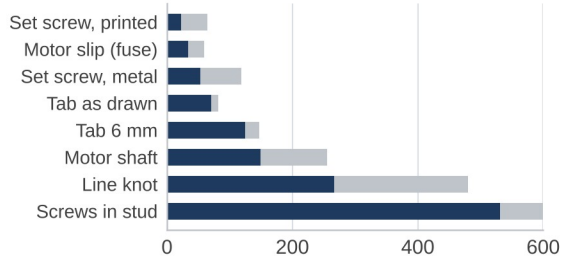
Need: 16.2 N worst cable for jeans; 34 N empty, 25 N full (calculated)

Winches can hold vs must hold (kg)



Need: a 0.45 kg claw; 0.86 estimated (CAD), 0.60 weighed; jeans 1.50 vs 1.56 (4 %)

Tension at which each part gives (N)



Need: the motor gives first (34-59 N); the printed set screw may not, so: metal insert

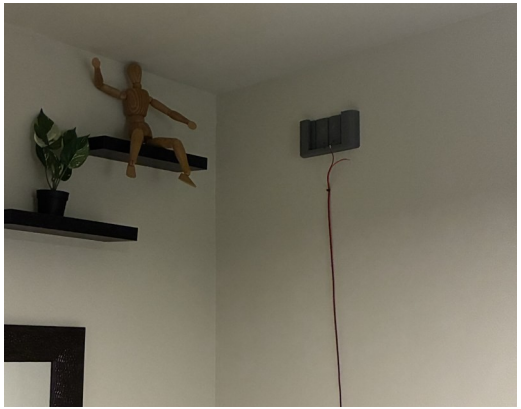
The factors

- Pull rises with current; heat rises with current squared
- 1.0 to 1.5 A: 50 % more pull for 2.25 x the heat
- Every gram on the claw is a gram of payload
- The pull test and the hold test are still planned

The mount assembly: slide-in on a dovetail

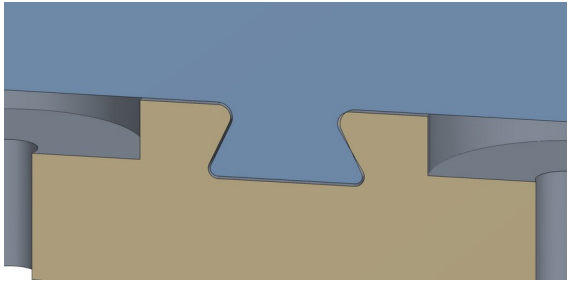
Slide-in mount and load path

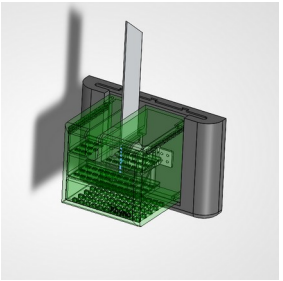
- The wall mount screws into a stud; the station slides onto it on a dovetail
- Dovetail fit: 0.175 mm a side, from fit coupons (measured)
- Load path: cable, spool, set screw, motor, tab, mount, screws, stud
- The motor slips at 34-59 N before anything breaks: the designed fuse (calculated)
- The spool encoder sees the slip: tracking fault, every station holds





The mount alone, before the station slides in; the station on the mount





Rib in the dovetail groove (CAD); the station with its shell transparent (CAD)

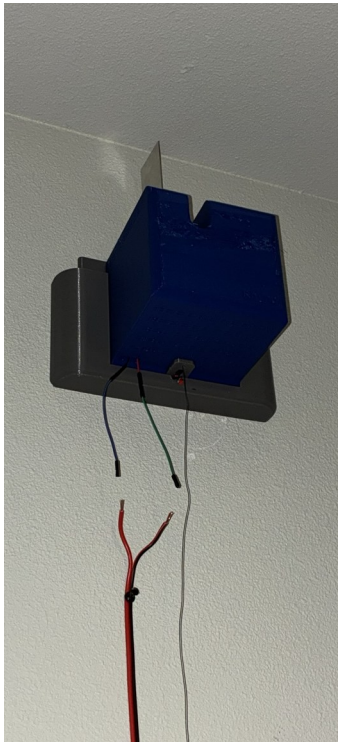
The mount assembly: fin, vents and the wall

Fin, vents, stud

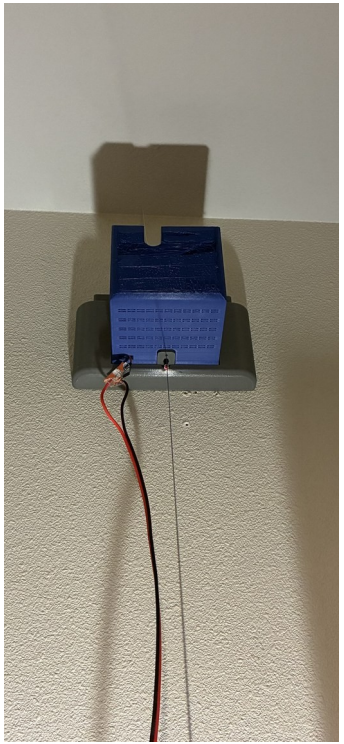
- The fin: aluminium on the motor face, out through a slot in the top of the shell
- Vents in the shell; vented shell + fin: 56-59 °C at 1.5 A (calculated)
- First station in a stud on the TV wall, 12 V down the wall
- The slide-in binds on insertion today; a 1° draft on the dovetail is the next iteration



On the TV wall, 12 V leads down



From below: fin through the slot



The line leaves the switch tab straight down

The claw, and why it is modular

The claw today

- Four-bar hands: within 1° of parallel over the 53 mm opening (calculated)
- A camera bay in the electronics housing; the camera looks down the tube
- One 7.4 V two-cell pack: 6 V regulator for the servo, 5 V buck for the board
- Weighed 0.60 kg with battery against a 0.45 kg budget

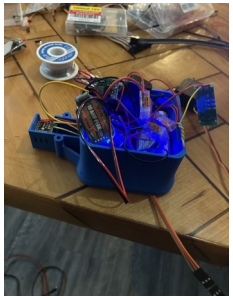
Modular, on purpose

- The gripper attaches at the tube, so other grippers can be tried
- A hard pinch was expected to grip better; testing showed alignment problems
- The first rack-and-pinion jaws left a 2 mm gap on thin cloth (Sep 30)
- TPU and other pad materials are being tried; still testing
- The v6 clamp was 4.6 x its mass budget in CAD: mass is tracked per part since









Assembled; the hands (CAD); the v6 clamp (CAD); the housing open

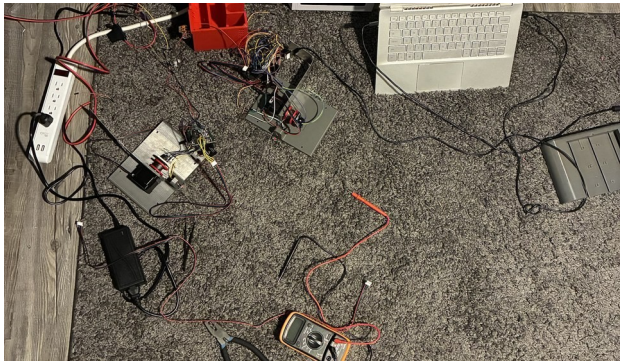
Electrical and testing: on the bench

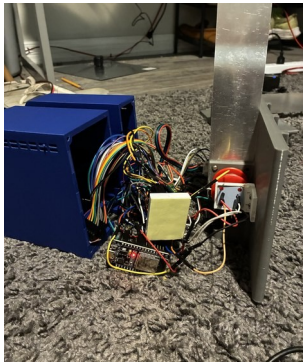
Station bench

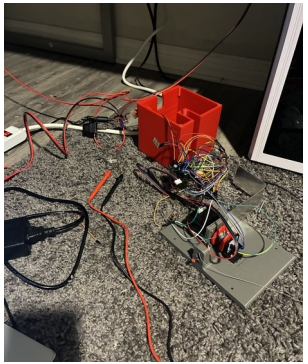
- Driver off at boot: a reset cannot move a motor
- Switch logic: LOW at rest, HIGH pressed or with a broken wire
- Encoder answers, magnet detected
- One spool turn within 60 counts: PASS on both boards (measured)
- A motor that logged steps but did not turn: logic rail at 1.14 V, not 3.3 V; VDD now soldered and measured first

Claw bench

- Live view from the claw camera on the phone
- Battery 7.2 V read through the divider, matching the meter (measured)
- The servo only moved once it had its own hardware timer; the camera owned the first
- Still open: servo end points unmeasured; camera not yet calibrated





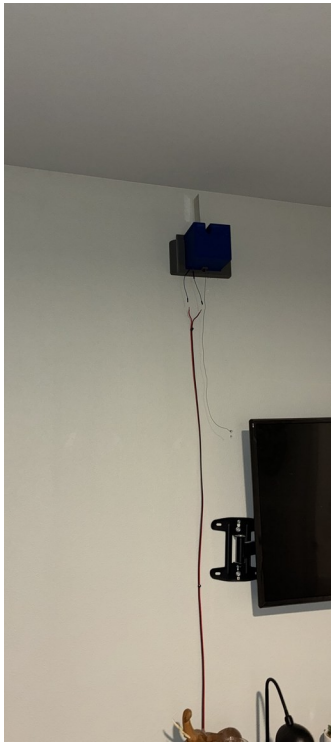


Two stations on the bench; a dev board and spool; the red shell with its fuse

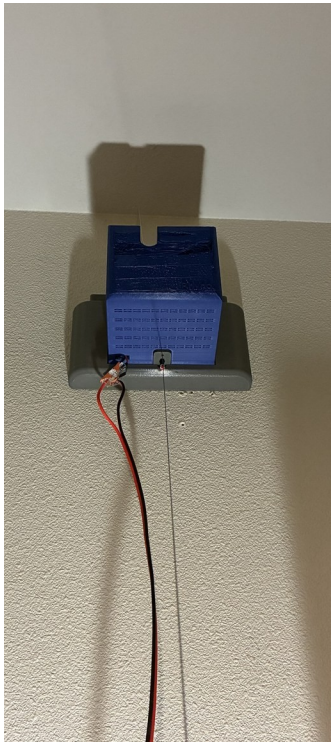
Electrical and testing: on the wall

Making it work on the wall

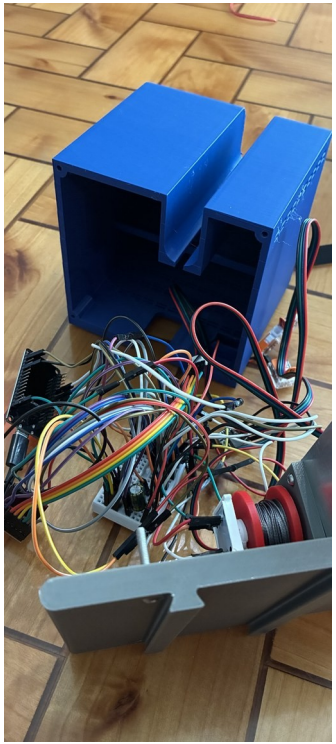
- Wi-Fi auto-join: a boxed station boots and joins the network by itself
- OTA: firmware updates arrive over Wi-Fi; no USB cable after boxing
- The motor stays off until the phone arms it
- A station accepts an update only while disarmed
- Still planned: pull test, hold test, first three-station lift



The first station on the wall



From below, line down



A station open: board on the bench

Startup and self-calibration: six states before the robot is READY

1 SELF_CHECK

- Claw online, battery above minimum
- Camera confirms the claw upright at the known height on the floor board
- Stations healthy, homing switches closed; arm

2 SLACK_TAKEUP

- One cable at a time reeled in until the camera sees the claw slide or tip (onset)
- Records spool position and slide bearing; backs off
- Bearings checked against the anchor layout; first fit of cable lengths

3 INITIAL_LIFT

- Pretension all cables; lift in small packets
- Abort on lateral drift, tilt, or commanded lift exceeding observed rise (slack cable)

4 VISUAL_INITIALIZATION

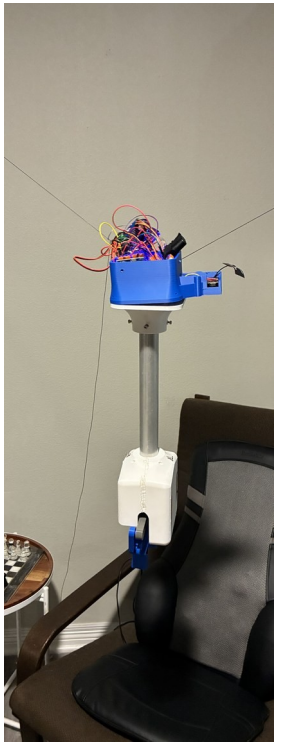
- Settled frame batch of the floor board
- Refit cable lengths and the room-to-marker transform

5 CALIBRATION

- Local → intermediate → expanded pose sets chosen by D-optimal design; refit between stages
- Held-out witness poses
- Accept on witness residuals, identifiability, predicted length sigma over the work volume

6 REFERENCE_VERIFICATION

- Optional bead references
- Trust level VERIFIED or LOCAL_ONLY from the margin check



Suspended, before CALIBRATION

No automatic retry; any station reboot forces the full sequence again.

Next iteration: what was not working, and what changes

3:1 printed gear stage

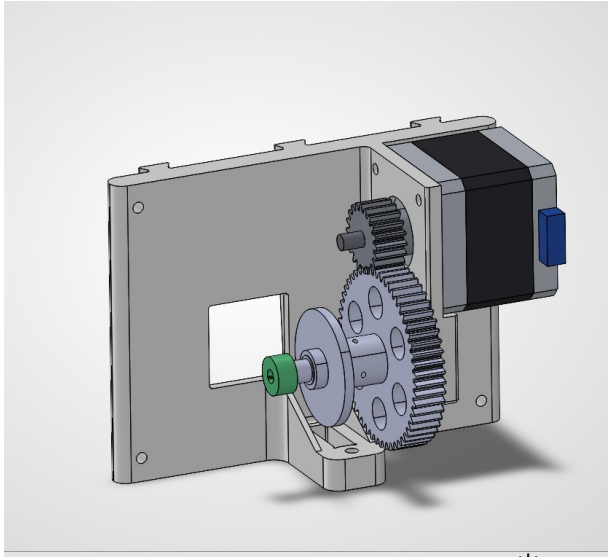
- 18T / 54T spur, printed PETG pinion
- Designed and checked in CAD; pinion printed; not fitted
- 2.7 x pull at 1.0 A, half the heat (calculated)
- Slip fuse moves to about 130 N (calculated)

1° draft on the dovetail

- Today the slide-in binds on insertion
- Thicker at the top, thinning toward the bottom
- The station self-locates as it slides in (planned)

Lighter back mount

- Today: about 200 x the motor's pull, 597 cm^3 solid
- Thinner walls and ribs: about 200 cm^3 (planned)
- Less print time and filament for anyone building it



Next iteration (CAD, 10/27): 3:1 spur stage from the motor pinion to the spool gear; spool on its own shaft, encoder magnet cap in green

Not working yet / still planned

- Pull test at 1.5 A: the measured limit goes into the planner (planned)
- Hold test: 90 minutes at 1.5 A, motor below 60 °C (planned)
- First three-station lift: self-check, calibration, a short move with 1 lb (planned)

Thank you

Jasper Buntinx | Mechanical Engineering | Texas A&M